
Ordinary Differential Equations

Kalaap

Part of a series of notes available at
kalaapks.github.io/notes.html

See kalaapks.github.io/notesDE.html for the notes on DEs.

Last updated: September 21, 2026

Contents

1	THE	
	THEORY OF	
	ODEs	3
1.1	System of ODEs	3
1.1.1	Visualizing ODEs	6
1.1.2	Ideas to solve the IVP	14
1.2	The linear ODEs	17
1.2.1	Visualizing linear ODEs	17
A		18
A.1	Python Codes	18
A.2	Geometry in Topology	18
A.3	Analysis	18
A.4	Method of Isoclines	18

List of Figures

1.1	Notice there are no abrupt changes in the flow of vectors arranged in the slope field because the fucntion $f(t, x) = t - x$ is 'nice' and that need not always be the case.	7
1.2	The 12 points from table	7
1.3	Notice there are abrupt changes in the flow of vectors here unlike last time since the fucntion $-t/x$ is not 'nice'.	8
1.4	Positive branch is choosen.	9
1.5	How the slope field will look completely depends on the value of λ . If $ \lambda \geq 0$ then growth/decay will be steeper.	10
1.6	Phase space of $x'' + 2\zeta\omega x' + \omega^2 x = 0$	11
1.7	Vectors in (a) are poking forward by unit 1 though because of the time component where as in (b) the time coordinate is killed.	12
1.8		13

THE THEORY OF ODEs

1

An ordinary differential equation is often introduced as an equation to be solved: given a relation involving an unknown function and its derivatives, our task is to find that function. But an ODE is much more than an equation waiting for an algebraic solution. At its heart, an ODE prescribes how a state changes. Given something like

$$X'(t) = f(t, X(t))$$

where the function f assigns a direction of motion to every state, we try to find a solution; a solution is a curve that follows these prescribed directions. This immediately leads to questions that are more fundamental than simply how to solve the equation:

- Does a solution exist? What does the curve look like?
- Is it unique?

We will approach these questions geometrically first, and then develop the analytical and computational viewpoints needed to understand ODEs.

1.1 System of ODEs

Cosnider a simple differential equation $\frac{dx}{dt} = a(t)x(t)$ that we have seen before. Solving this naively will lead us to cross multiplying dt and then taking the integral on both sides of the equation $\frac{dx}{x(t)} = a(t)dt$. Firstly, this is not entirely correct since dt is not a scalar but

$\lim_{t \rightarrow 0}$ instead, hence, we can't cross multiply. Let's for a moment not worry about the technical reasonings behind why the above method is still allowed or the precise domain of integration and only focus on what we get as the solution.

$$\int \frac{1}{x} dx = \int a(t) dt$$

$$x := x(t) = e^{\int a(t) dt} \quad (1.1)$$

If $t \in I$ (some interval in $\mathbb{R}$) then our solution $x(t)$ can be viewed as a graph in (t, x) -plane. at any point $(t, x(t))$ the curve shows slope that equals $a(t)x(t)$ since that is exactly what $\frac{dx}{dt} = a(t)x(t)$ means. That is the entire point of an ODE, **given the slope at every point can we find the curve that has the prescribed slope?**

Naturally things get complicated. Let us point out some of the complications and mention which among them we will try to tackle right now.

- In general our ODE is not as nice as the one we saw. We have

$$\frac{dx}{dt} = f(t, x(t)) \quad (1.2)$$

As before we still have the same interpretation; once we do successfully find a curve $x(t)$ it will have slope $f(t, x(t))$ at every $(t, x(t))$. **The cross multiplying technique won't work anymore obviously** and hence things have started to get messy.

- Things can get even messier. Understanding this will require us to look at an ODE from a different perspective. What we saw up till now were 'given ODE, solve for the curve' problems but we could very well have a 'given a curve, what should the governing ODE be' problem. For example consider the function $x(t) = \cos(t) + \sin(t)$. We do not get an ODE by taking the derivative once and we will need to use the 2nd derivative too to get a relation that connects back to $x(t)$.

$$x'(t) = -\sin(t) + \cos(t)$$

$$x''(t) = -(\cos(t) + \sin(t)) = -x(t)$$

This naturally makes sense when you think of a physical system where $x(t)$ represents position and $x'(t)$ represents velocity. To get the precise trajectory we do sometime need to know the accelerations too. Consider the case when the acceleration is 0, that is,

$$x''(t) = 0$$

Then, $x(t) = mt + c$

Acceleration is what changes velocity. If acceleration is zero, velocity never changes. If velocity never changes, the particle keeps moving in exactly the same direction at exactly the same speed hence a straight line. The point is **we can have curves that need 2 or higher order of derivatives to be represented as a ODE.**

- Now that we understand we have to allow higher order derivatives why not consider things such as $(x'x)^2 = x''$ or $\frac{x''x^3}{x'''} = (x')^2$? surely even they represent some trajectory of a particle that moves really really weird! That is a valid proposal and we do have them under what we call **"nonlinear ODEs"**. Right now, we will start with the "linear" kind and then move onto the nonlinear kind. Let's make the distinction precise.

Definition 1.1. A general linear ODE of order n with dependent variable x and independent variable t has the form

$$x^{(n)}(t) + a_{n-1}(t)x^{(n-1)}(t) + \cdots + a_1(t)x'(t) + a_0(t)x(t) = g(t) \quad (1.3)$$

where coefficients $\{a_j\}_{j=0,\dots,n-1}$ are functions of t .

Note that both the explicit ODEs we saw, namely $x'(t) = a(t)x(t)$ and $x''(t) + x(t) = 0$, are of this form. We introduce dummy variables $\{x_1, \dots, x_n\}$ and define them to be,

$$\begin{aligned} x_1(t) &= x(t) \\ x_2(t) &= x'(t) = x'_1(t) \\ &\vdots \\ x_n(t) &= x^{(n-1)}(t) \end{aligned}$$

We can now turn this n -order linear ODE into a linear system of n -many 1-order ODEs. Let $X(t)$ be the vector $(x_1(t), \dots, x_n(t))^T$. Then, 1.1 has the form:

$$X'(t) = \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \\ \vdots \\ x'_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_0(t) & -a_1(t) & -a_2(t) & \cdots & -a_{n-1}(t) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ g(t) \end{bmatrix} \quad (1.4)$$

So, exactly like the very first case we have $X'(t) = A(t)X(t) + G(t)$; here $X(t)$ will represent a point in $\mathbb{R}^N$ that is built of $sp_{\mathbb{R}}(x_1, \dots, x_n)$. We will hence put our entire **focus on studying the problem $X'(t) = f(t, X(t))$** .

[Question] Why did we have to shift from $A(t)X(t)+G(t)$ to a general $f(t, X(t))$?

The reason is any n -order linear ODE can be turned into a single system of n -many first order ODEs (as done above) but the converse isn't true. **Not every $X'(t) = f(t, X(t))$ represents n -order linear ODE.**

- We can have **coupled linear systems** where the variables $\{x_1, \dots, x_n\}$ are not independent of one another resulting in a more complicated matrix than the one we saw above. One such example will be considering two n order linear ODEs and then turn them into a single matrix problem. We will end up with a $2n$ components long vector $X(t)$ and a matrix $A(t)$ that has two blocks of what we had above. Consider,

$$\begin{cases} x^{(n)}(t) + a_{n-1}(t)x^{(n-1)}(t) + \cdots + a_1(t)x'(t) + a_0(t)x(t) = 0 \\ y^{(n)}(t) + b_{n-1}(t)y^{(n-1)}(t) + \cdots + b_1(t)y'(t) + b_0(t)y(t) = 0 \end{cases}$$

This will give us:

$$Z'(t) = \begin{bmatrix} X'(t) \\ Y'(t) \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} X(t) \\ Y(t) \end{bmatrix} \quad (1.5)$$

This is an example of linear system of ODEs that does not look as simple as 1.4. So, we need to relax the condition on structure of A ; later on when we give explicit solution of linear system of ODEs we will let A be any $n \times n$ matrix.

• **Non-linear ODEs** have the $X' = f$ form as well. Recall the non linear ODEs we saw before, $(x'x)^2 = x''$ and $\frac{x''x^3}{x'''} = (x')^2$. Let's turn one of them into our matrix system. Define

$$x_1 = x, \quad x_2 = x', \quad x_3 = x''.$$

From our equation,

$$\frac{x''x^3}{x'''} = (x')^2$$

we get

$$x''' = \frac{x''x^3}{(x')^2} \implies x'_3 = \frac{x_3x_1^3}{x_2^2}.$$

So the whole thing becomes

$$X'(t) = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ \frac{x_3x_1^3}{x_2^2} \end{bmatrix}$$

which looks like the general form $X' = f(t, X)$. Hence, we will tackle with the general problem $X'(t) = f(t, X(t))$ directly which encompasses the n-order linear ODEs and the nonlinear ODEs.

1.1.1 Visualizing ODEs

This is very essential to everything we see next section onwards. Before attempting to formulate ways for finding solutions, we will try to make the geometry of $X'(t) = f(t, X(t))$ more precise. We will see this in cases based on the dimension of the space where the graph of the solution curve lies.

Definition 1.2. Phase space : A phase space in the context of ordinary differential equations is a multi-dimensional space where each axis represents a state variable of a dynamical system, and every point in the space corresponds to a complete possible state of that system. We are using the variable $X = (x_1, \dots, x_n)$ to represent them.

When $X = (x_1, x'_1, x''_1, \dots)$, a point in Phase space will mean (*position, velocity, acceleration, ...*). BUT, mind you, X need not be $(x_1, x'_1, x''_1, \dots)$. **A system of first-order ODEs can also represent entirely different physical quantities, such as three separate spatial variables (x_1, x_2, x_3) capturing the motion of a particle in 3D space and this makes CASE 3 discussion important.** The cases will be, hence, built on the dimension of phase space.

• **CASE 1 :** Here, $(t, X(t)) = (t, x(t)) \in \mathbb{R} \times \mathbb{R}$.

This is basically the simplest case and we no systems yet.

Example 1.3. $x'(t) = f(t, x) := t - x$

t	1	0	-1	0	1	2	0	-2	0
x	0	1	0	-1	-1	0	2	0	-2
$x'(t)$	1	-1	-1	1	2	2	-2	-2	2

With this table we can proceed to plot it now. At all (t, x) we plot the value of $f(t, x)$. This is like imagining the graph of $f(t, x)$; we will not do that however. **Instead of looking at the assigned value of f at (t, x) we will imagine a small vector at (t, x) with slope $f(t, x)$.** We

can absolutely choose to do this, since, how we want to visualize something is upto us. For example, at $(1, 0)$, $(0, 1)$ and $(1, -1)$ we will imagine small vectors with slope 1, -1 and 2.

We see geometrically in 1.1, the trajectory is very clear. In case you are plotting it by hands (which a perfectly good way to start and how i started too - the codes come later), there is an easier way to do this which is called the method of isoclines but we won't see it now because that will break the flow of what we are building towards. See A.4 (it is suggested to see it after this section is done and not right now).

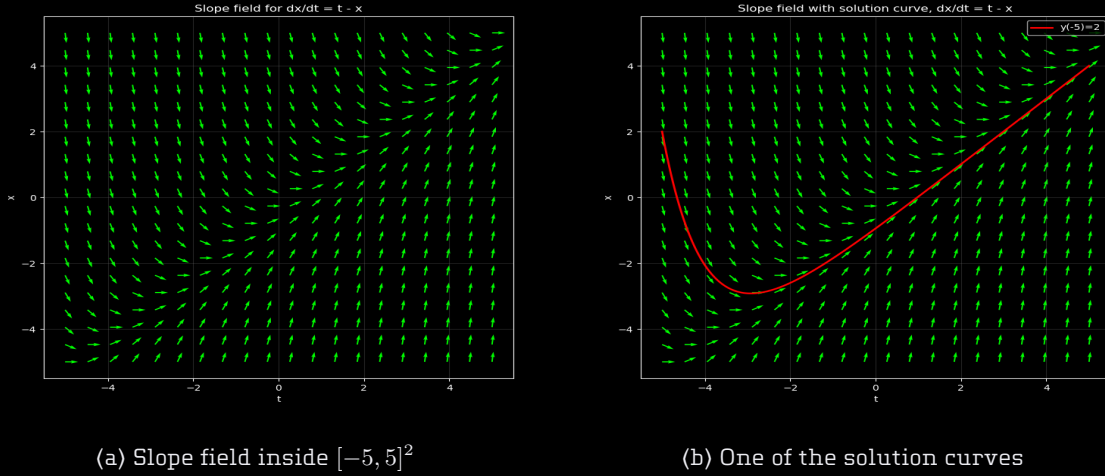


Figure 1.1: Notice there are no abrupt changes in the flow of vectors arranged in the slope field because the function $f(t, x) = t - x$ is 'nice' and that need not always be the case.

Example 1.4. $x'(t) = f(t, x) := -\frac{t}{x}$

t	1	0	-1	0	2	0	-2	0	$\sqrt{8}$	$-\sqrt{8}$	$-\sqrt{8}$	$\sqrt{8}$
x	0	1	0	-1	0	2	0	-2	$\sqrt{8}$	$\sqrt{8}$	$-\sqrt{8}$	$-\sqrt{8}$
$x'(t)$	∞	0	∞	0	∞	0	∞	0	-1	1	-1	1

Like before we can plot it now; let us try to do it by hands.

We get the figure 1.2 but this is not quite right though!

Think analytically:

For fixed t , $f(t, x) = tg(x)$ and $g(x) = -\frac{1}{x}$.

Focus on g only.

$$\lim_{x \rightarrow 0^+} g(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} g(x) = \infty$$

So the ∞ we see in the table needs to be taken with a sign since that will dictate the direction of the vector. The vectors slightly above $(1, 0)$ are due to the $\{x \geq 0\}$ case and hence they should point down. Similarly, the vectors slightly below $(1, 0)$ are due to the $\{x \leq 0\}$ case and hence they should point up. Let us see the computer generated plots now. As stated, the vectors in $\{x \geq 0\}$ of

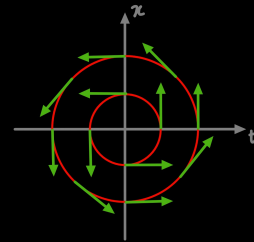
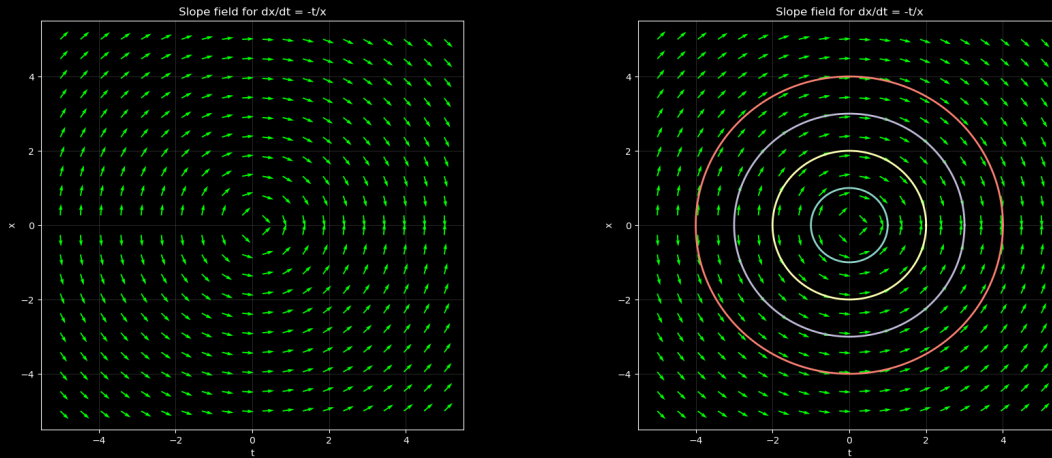


Figure 1.2: The 12 points from table

Figure 1.3a do not match with what we had drawn by hands.

This is basically due to how the function $f(t, x) = -\frac{t}{x}$ behaves on $\{x = 0\}$.

Remark 1.5. True solution $x(t) = \pm\sqrt{R^2 - t^2}$ of the above ODE are circles. So, the solution curves $x(t)$ are not necessarily functions globally. However, they are functions locally (due to Implicit function theorem, see A.3) and that is the reason why we will work in a local nbds while building the entire theory of solutions in Section 1.1.2



(a) Slope field inside $[-5, 5]^2$

(b) Solution curves at $R = 1, 2, 3, 4$

Figure 1.3: Notice there are abrupt changes in the flow of vectors here unlike last time since the function $-t/x$ is not 'nice'.

[Question] In the graph of the solution curves, the paths even go back in time! How is that fine for a "trajectory of a particle"?

This problem is due to the fact that we are seeing 'time' as an actual axis but in reality we do not have that. To see the true path of the particle, we have to see 'the movement' only in the position-subspace, that is, the x -axis here. The particle is moving in one-dimension.

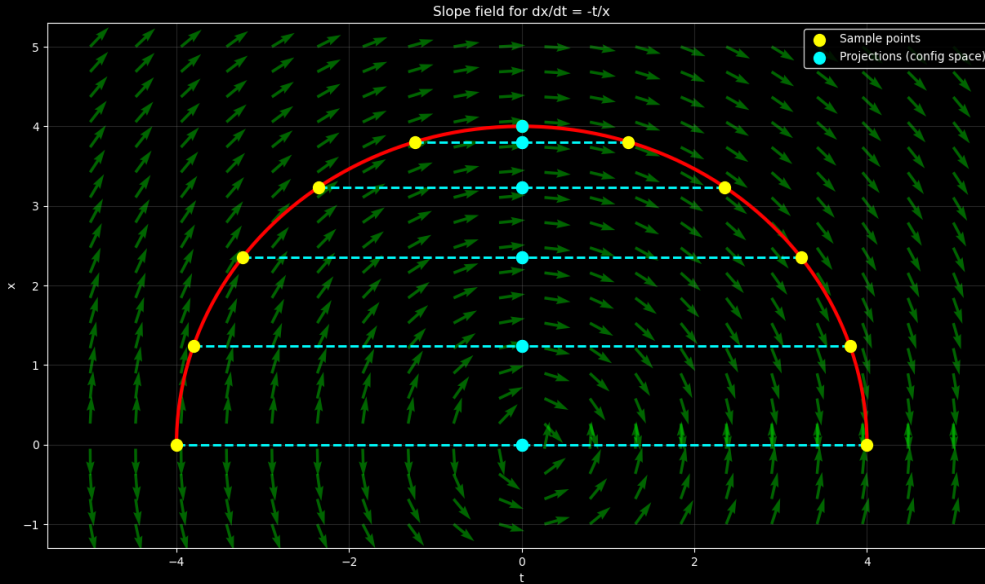


Figure 1.4: Positive branch is chosen.

Solution $x(t) = \pm\sqrt{R^2 - t^2}$ makes sense only for $t \in [-R, R]$.

We have to start with either the positive branch or the negative branch; that choice will basically tell us we are on the upper half plane or the lower half plane of the Phase plane.

If we now choose to work with $x(t) = \sqrt{R^2 - t^2}$ and allow t to vary in $[-R, R]$, at $t = -R$ we will be at $(-R, 0)$. As, $t \rightarrow 0$, our position moves from $(-R, 0)$ to $(0, R)$ gradually. Then, as $t \rightarrow R$, our position moves from $(0, R)$ back to $(R, 0)$. So, the yellow points are our trajectory in abstract Phase space and the cyan points are our actual path of the particle.

Also, mathematically we had $t \in [-R, R]$ because beyond that it becomes complex numbers. In our physical model, this problem is visible too - we literally don't know where the particle will go beyond R since $(R, 0)$ wants to go down but the phase plane is pushing it up! It is stuck.

Definition 1.6. Configuration Space : This is the spatial subspace of Phase space in which the particle physically moves around.

In the example we just saw, Phase space is same as the configuration space. The distinction will make sense when we get to these Questions because right now, we do not have enough understanding.

Example 1.7. $x'(t) = \lambda x(t)$, $\lambda \in \mathbb{R}$

The true family of solution curves is the exponential family $\{ce^{\lambda t} : t \in \mathbb{R}\}$ as visible in Figure 1.5. This particular ODE is going to be one of the most important one for us. Note that, the t variable is not appearing explicitly on right; so $f(t, x)$ is constant in time. So, for anyone drawing by hands we can simply draw the slope field along the vertical line $t = 1$ and then copy it on every t .

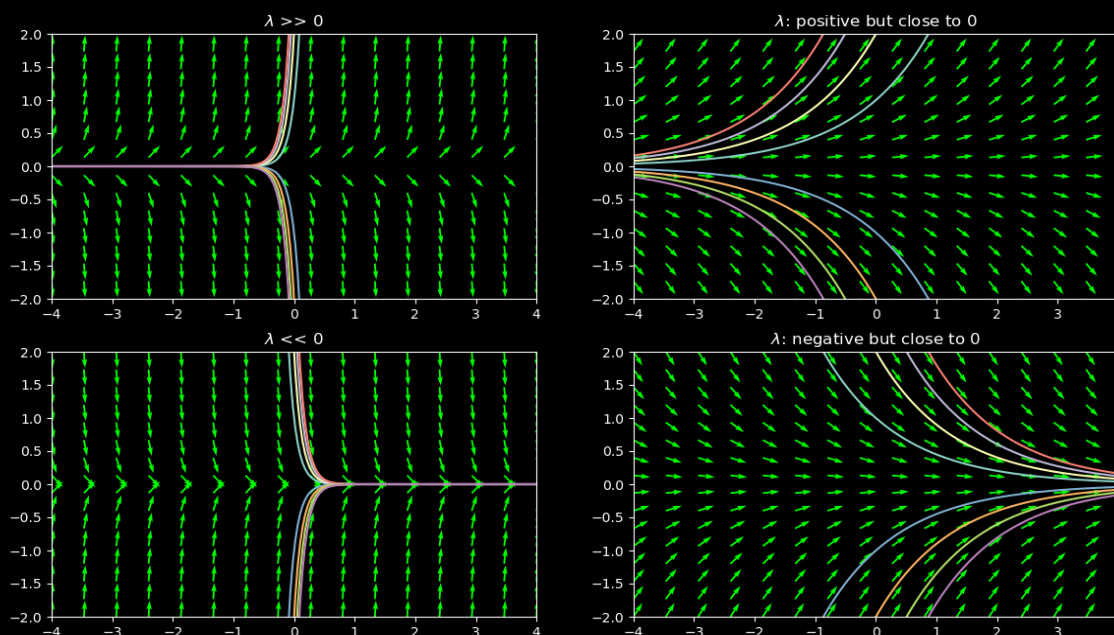


Figure 1.5: How the slope field will look completely depends on the value of λ . If $|\lambda| \geq 0$ then growth/decay will be steeper.

As before, if you want to see the path of the particle that shows $x' = x$ motion we only look at the movement in the configuration space, that is on x -axis. This is $\lambda = 1$ case. It starts moving slowly from somewhere above 0 and goes towards ∞ (or $-\infty$ if we start somewhere below 0) with velocity that gets faster and faster as we move away from 0. If you ponder, this is exactly how one would read $x' = x$ in words.

Real-life models where this shows up include radioactive decay, population growth, etc.

Remark 1.8. Notice one of the most crucial difference between the last two examples and this one we just saw. The first two were not constant in time but this is. It is very important to realise, if we look at the plot one t at a time, for every time t we may not/may different map. *This observation might not feel so obvious when we move to higher dimensions and each time slices are themselves 2-D or 3-D vector fields.* We will see this in the following example.

- **CASE 2:** Here, $(t, X(t)) = (t, x_1(t), x_2(t)) \in \mathbb{R} \times \mathbb{R}^2$.

This is the simplest case of a system and since we maxed out on the available dimensions we cant visualize the slope fields **CASE 3** onwards. This is very important for us since in Physics most of the phenomenons we try to model are $2^n d$ order systems like Newton's Laws of motion. So, we will almost always use the $(t, x_1, x_2) = (t, x_1, x'_1)$ space.

Example 1.9. $x'' + 2\zeta\omega x' + \omega^2 x = 0$

This is the equation of a **damped harmonic oscillator**. Duhhh! everyone of you math folks know this, right? Even a physicist does!

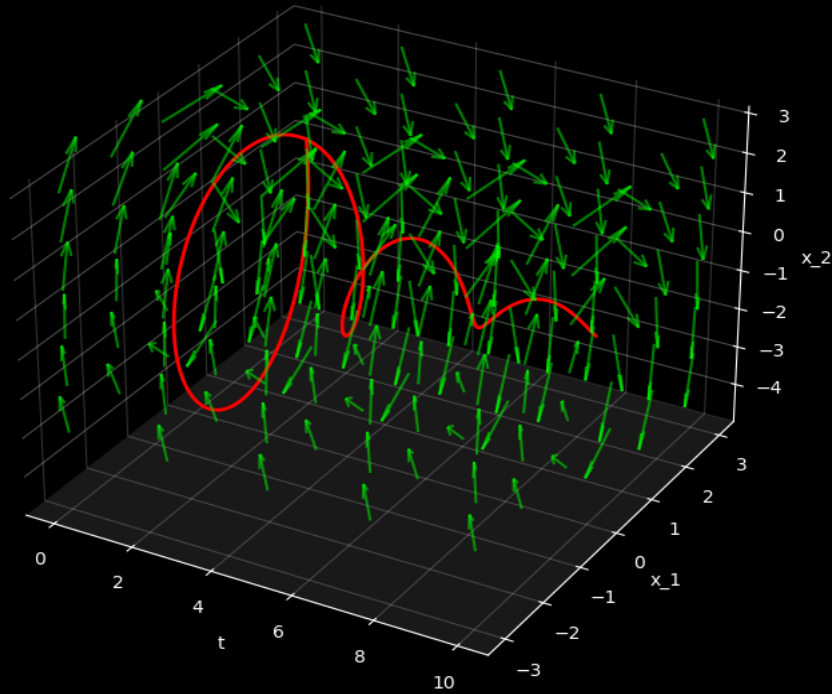


Figure 1.6: Phase space of $x'' + 2\zeta\omega x' + \omega^2 x = 0$

We turn it into a system so that we can plot it.

$$\frac{d}{dt} \begin{bmatrix} t \\ X(t) \end{bmatrix} = \begin{bmatrix} t' \\ x'_1 \\ x'_2 \end{bmatrix} = \begin{bmatrix} 1 \\ x_2 \\ -2\zeta\omega x_2 - \omega^2 x_1 \end{bmatrix} \quad (1.6)$$

We plotted this vector field in 3-D and got the figure 1.6 ; and having seen the plot, I should mention that it is about time to stop drawing the vector fields by hand. This looks so complicated that we can not even make sense of what the directions are without the solution curve.

There is a hidden geometry though. It is not at all obvious from the 1.6 plot but these vector fields are constant on each $\{t = c\}$ -plane. This is clear from the equation 1.6 itself since the vector on the right does not have time variable ' t ' showing up. On every time slice we have the same vector field. So, in such scenerios it is enough to focus on the 2-D vector field itself. The trajectory in this 2-D vector field will be that of a inward spiral and then in the full space (with time axis) due to the ' t ' component we should move forward in time with constant velocity 1. This gives us the forward moving inward spiral - exactly as we have gotten in the figure 1.7a.

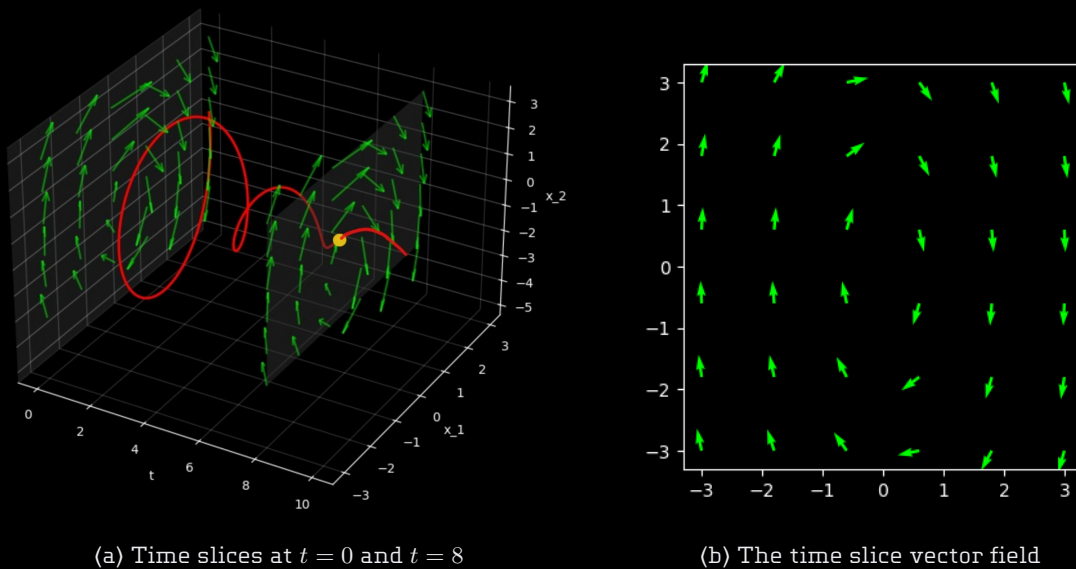


Figure 1.7: Vectors in (a) are poking forward by unit 1 though because of the time component where as in (b) the time coordinate is killed.

This brings us to the important conclusion that **when the time variable doesn't appear explicitly in the system we can simply work in a vector space of lower dimension.** The cases where these reduction can be done are called Autonomous systems. This is not a small conclusion, reducing the dimensionality reduces the number of machine calculations greatly.

Definition 1.10. Autonomous ODEs : An autonomous ODE does not explicitly depend on the independent variable (like time), whereas a non-autonomous ODE depends explicitly on both the dependent and independent variables.

This idea of 'time-slicing' is nice even when the system is non-autonomous. For such systems, all that changes is the vector fields in each of the time slices are now different. This is what we mentioned earlier in Remark 1.8.

This gives us a way to visualize f as a family of lower dimensional vector fields. In Case 1, we saw f taking points from 2-D to $\mathbb{R}$. In Case 2, f is a map that is taking (t, x_1, x_2) to $f(t, x_1, x_2)$, that is, takes a point from 3-D to a point in $\mathbb{R}$. **We will always have f to be a function from $(1 + n)$ -D space down to n -D space because**

$$f(\text{time} + \text{State variables}) = (\text{State variables})'$$

Since it is not possible to look at this as a vector field, we will adopt the 'slicing method' for seeing f at fixed times t . In doing so we will be able to look at 'vector fields' such that at every point in X we have vector X' attached.

The mental picture I like is f being associated with a 'flow' - like a vector field that changes with time. This is the reason you might see in some places the notation $f_t(x)$ for $f(t, x)$.

ADD picture (4-5 time slices with different vector fields)

Definition 1.11. Static Extended Phase space : If we stack all of these n -D time slices on top of each other chronologically, we build an $(n + 1)$ -dimensional space called the Extended Phase Space with coordinates $(t, X) = (t, x_1, x_2, \dots, x_n)$ and with the vector $(1, X') = (1, x'_1, x'_2, \dots, x'_n)$ attached at that coordinate.

Remark 1.12. What we were plotting in Case 1 was, hence, the Static Extended Phase space and not the Phase space. The point (t, x) had the vector $(1, x')$ attached.

In this extended space, the vector field is completely static because time has been locked down as a physical spatial axis. Every vector points slightly "forward" along the time axis because $\frac{dt}{dt} = 1$. For non-autonomous systems, we need the full dynamic of extended Phase space. We will hence build our theory entirely around finding a curve in this extended space.

Example 1.13. $x'' + \mu x' + \frac{g}{L} \sin(x) = 0$

This is the equation of an **Oscillating pendulum** and is surprisingly close to the previous example of damped harmonic oscillator equation $x'' + 2\zeta\omega x' + \omega^2 x = 0$. If we don't bother with the constants, the only change is how one has x where as the other has $\sin(x)$. This is, unfortunately, not a small change! The appearance of *sinusoidal* function makes the current equation a nonlinear one and the non linear ones often times don't have closed form solutions as is the case with this one.

• **CASE 3 :** Here, $(t, X(t)) = (t, x_1(t), x_2(t), x_3(t)) \in \mathbb{R} \times \mathbb{R}^3$.

We have built enough understanding now and we no longer need to see explicit examples to get a feel for what is going to happen. As mentioned, we can't see the slope field anymore and will only see the phase space $\mathbb{R}^3$. In particular, let's see a small subset S of $\mathbb{R}^3$.

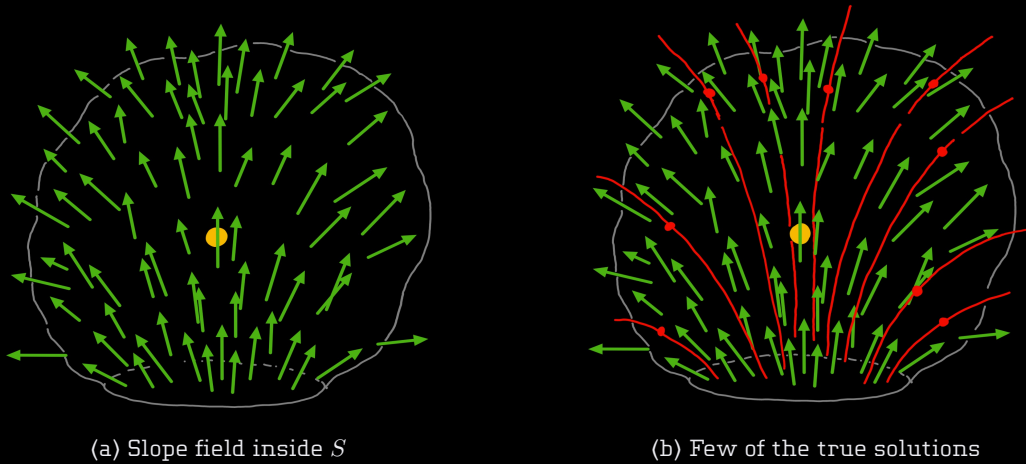


Figure 1.8

[Question] Under which case does these models fall?

- **a particle moving in 2-D:** The particle p has two spatial dimensions (p_x, p_y) and since Law of motion is 2^{nd} order ODE, we have the velocities $\{p'_x, p'_y\}$ for both directions; (p'_x, p'_y) tells us where p is going in the plane.

So, the extended phase space $(t, X(t))$ is $(t, p_x, p_y, p'_x, p'_y)$, the phase space (p_x, p_y, p'_x, p'_y) is 4-D and the configuration space is (p_x, p_y) .

- **a particle moving in 3-D:** Similar to the above case, we will have 3 spatial dimensions for the particle p . So, $(t, X(t)) = (t, p_x, p_y, p_z, p'_x, p'_y, p'_z)$ is the extended phase space. The phase space $(p_x, p_y, p_z, p'_x, p'_y, p'_z)$ is 6-D and the configuration space is (p_x, p_y, p_z) .

- **2 particle moving in 3-D?** The phase space will be 12-D. Let the particles be p and q . Then, $(t, X(t)) = (t, \underbrace{p_x, p_y, p_z, p'_x, p'_y, p'_z}_{\text{position of p}}, \underbrace{q_x, q_y, q_z, q'_x, q'_y, q'_z}_{\text{position of q}})$ is the extended phase space.

Configuration space is $(p_x, p_y, p_z, q_x, q_y, q_z)$. This is the standard set up when we try to study the movement of two planetary objects in space. How two bodies will move in 3-D space under eachother's influence boils down to get the path of p and q in configuration space.

- **3 particle moving in 3-D?** Let the particles be p, q & r . The extended phase space is 19-D. This is the standard setup of how three planetary bodies move in space. This is the famous **3 Body problem** which is so chaotic that it can't be solved! This means given the three initial positions p_0, q_0, z_0 in $\mathbb{R}^3$ and given their initial velocities we can't say for ceratin where the three bodies will be after some time has passed.

ADD picture

The need for considering a n -D problem is hence clear. If we can find a valid solution or not is the problem we want to tackle now.

1.1.2 Ideas to solve the IVP

Our job will always be to find a curve X in n -dim (axes are $x, x', \dots, x^{(n-1)}$) that moves along the slope field (or phase space if we are choosing to not have the time axis); also, as established in Remark 1.5, we will only be interested in local solutions inside a subset of $\mathbb{R} \times \mathbb{R}^n$, namely D .

Definition 1.14. Solution : Let $n \in \mathbb{N}$, $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and let $f : D \rightarrow \mathbb{R}^n$. Then, $X : I \rightarrow \mathbb{R}^n$ where I is an open interval of $\mathbb{R}$ is said to be a solution of ODE

$$X'(t) = f(t, X(t))$$

if:

1. $X : I \rightarrow \mathbb{R}^n$ is differentiable.
2. $(t, X(t)) \in D, \forall t \in I$
3. $X'(t) = f(t, X(t)), \forall t \in I$.

Definition 1.15. Initial value problem : Let $n \in \mathbb{N}$, $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be open, $f : D \rightarrow \mathbb{R}^n$ and $(t_0, X_0) \in D$. We say that the initial value problem (IVP)

$$\begin{cases} X'(t) = f(t, X(t)) \\ X(t_0) = X_0 \end{cases}$$

has a solution if $\exists$ open interval $I \subseteq \mathbb{R}$ and $X : I \rightarrow \mathbb{R}^n$ such that

1. $t_0 \in I$
2. $X : I \rightarrow \mathbb{R}^n$ is differentiable
3. $(t, X(t)) \in D, \forall t \in I.$
4. $X'(t) = f(t, X(t)), \forall t \in I.$
5. $X(t_0) = X_0$

Definition 1.16. ε -approximate solution : Let $n \in \mathbb{N}$, $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $f \in \mathcal{C}(D, \mathbb{R}^n)$. Consider ODE $X'(t) = f(t, X(t))$ and let some $\varepsilon > 0$. Also consider some open interval I and some function $X_\varepsilon : I \rightarrow \mathbb{R}^n \ni X \in \mathcal{C}(I; \mathbb{R}^n)$. Then X_ε is ε -approximate solution if:

1. X_ε is $\mathcal{C}^1$ on I except for finitely many points.
2. $(t, X_\varepsilon(t)) \in D \forall t \in I.$
3. $\|X'_\varepsilon(t) - f(t, X_\varepsilon(t))\| < \varepsilon \quad \forall t \in I \setminus \text{finite set } S.$

These solution $X_\varepsilon(t)$ are basically the polygonal approximation of the true solution and the finite subset S of I that is mentioned in the theorem are the points t where the graph $(t, X_\varepsilon(t))$ forms the corners. This obvious idea is surprisingly powerful. At this preliminary stages, one can already guess that the limit $\lim_{\varepsilon \rightarrow 0} X_\varepsilon(t)$ might give us out true solution. In actual physical model set ups where we search for the solutions we actually look at these $X_\varepsilon(t)$ solution for small enough ε since sending ε to 0 is not an option for computers. There are, however, some practical issues that comes up when we try to even find the ε -approximate solutions. We explore these in Numerical Methods; for now we are in the realm of analysis where we can take limits.

The following lemma is not only an existence result but also gives an explicit construction of the ε -approximate solution. In numerical methods, we call this method the **Forward Euler/Explicit Euler** method. Let us define some special sets called 'cylinder'-sets in $\mathbb{R} \times \mathbb{R}^n$.

$$\begin{aligned}
 \mathfrak{C}(c; r) &:= (t_0 - c, t_0 + c) \times B(X_0; r) \\
 \mathfrak{C}(c; \bar{r}) &:= (t_0 - c, t_0 + c) \times B[X_0; r] \\
 \mathfrak{C}(\bar{c}; r) &:= [t_0 - c, t_0 + c] \times B(X_0; r) \\
 \mathfrak{C}(\bar{c}; \bar{r}) &:= [t_0 - c, t_0 + c] \times B[X_0; r]
 \end{aligned} \tag{1.7}$$

Remark 1.17. Geometry of the cylinder sets : They are a generalized name and they look like actual cylinders only in the special case when ball B is in $\mathbb{R}^2$ and we have the ℓ^2 norm in $\mathbb{R}^2$. The actual shape changes depending on how the base ball looks, that is, what ℓ^p norm is endowed. For example, when we have ℓ^∞ norm we get a cuboid (we are going to use this norm in the next proof) since the base ball is a square [see A.2].

We will be looking at the extended Phase space inside this cylinder and in the following lemma find a ε -approximate solution. Over time, we may get real far from true solution, as the picture suggests, but we are looking at local polygonal approximations only and hence we are fine.

ADD picture

Lemma 1.18. Existence of ε -approximate solution : Let $t_0 \in \mathbb{R}$ and $X_0 \in \mathbb{R}^n$; $c, r > 0$. Let $f \in \mathcal{C}(\mathfrak{C}(\bar{c}; \bar{r}); \mathbb{R}^n)$ and let $M > 0$ be such that

$$M \geq \sup\{\|f(t, x(t))\| : (t, x) \in \mathfrak{C}(\bar{c}; \bar{r})\}$$

Let $h := \min\{c, r/M\} \leq c$. Then for every $\varepsilon > 0$, $\exists$ an ε -approximate solution

$$X_\varepsilon \in \mathcal{C}([t_0 - c, t_0 + c], B[X_0; r])$$

in $[t_0 - h, t_0 + h]$ of $X'(t) = f(t, X(t))$ satisfying:

1. $X_\varepsilon(t_0) = X_0$
2. $\|X_\varepsilon(t) - X_\varepsilon(s)\| \leq M|t - s| \quad \forall t, s \in [t_0 - h, t_0 + h]$

Theorem 1.19. Cauchy-Peano : Let $n \in \mathbb{N}$, $t_0 \in \mathbb{R}$, $X_0 \in \mathbb{R}^n$; let $c, r > 0$ and $f \in \mathcal{C}(\mathfrak{C}(\bar{c}; \bar{r}); \mathbb{R}^n)$. $M \geq \sup\{\|f\|_C\}$. Let $h := \min\{c, r/M\}$.

Then $\exists X$ continuous on $[t_0 - h, t_0 + h]$ and continuously differentiable on $(t_0 - h, t_0 + h)$ satisfying IVP.

Proof: Sequence of ε -approximate $\sim \frac{1}{m}$ -approximate.

Lemma 1.20. Gronwall lemma : $a, b \in \mathbb{R}$, $c \in [a, b]$ and g continuous on $[a, b]$ $\ni 0 \leq g(t) \leq \alpha + \beta \left| \int_c^t g(s) ds \right|$ for some $\alpha, \beta > 0$. Then $g(t) \leq \alpha e^{\beta|t-c|}$

Definition 1.21. Successive approximation : Let $n \in \mathbb{N}$, $t_0 \in \mathbb{R}$, $X_0 \in \mathbb{R}^n$; $c, r > 0$. C as usual. $f \in \mathcal{C}(C, \mathbb{R}^n)$ and $M \geq \sup\{\|f\|_C\}$. Let $h := \min\{c, r/M\}$. Let us have some IVP. Then successive approximation is a sequence of integral functions continuous on $[t_0 - h, t_0 + h]$ defined as:

$$\begin{aligned} S_1(t) &= X_0 \quad \forall t \in [t_0 - h, t_0 + h] \\ &\vdots \\ S_{m+1}(t) &= X_0 + \int_{t_0}^t f(u, S_m(u)) du \quad \forall t \in [t_0 - h, t_0 + h]. \end{aligned}$$

1.2 The linear ODEs

Here we deal with and give a complete explicit solution of $X'(t) = A(t)X(t) + G(t)$ where A is a general $n \times n$ matrix. Note that here A is not of the kind we saw in 1.4. Once we provide the complete theory to solve these ODEs, 1.4 will become a special case of this and thus we would have solved all possible n-order linear ODEs.

1.2.1 Visualizing linear ODEs

A

- A.1 Python Codes
- A.2 Geometry in Topology
- A.3 Analysis
- A.4 Method of Isoclines